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Research Topic: External Sensory Competition and Symbol-Selection Latency on a Fixed AAC Grid in Autistic Adolescents

Environmental Sensory Competition and AAC Symbol Selection in Autistic Adolescents: A Pilot Study

Introduction and Problem Statement

High-tech AAC systems are a path to expressive language for autistic people with complex profiles. Most commercial systems use visual grid layouts. Picking just one symbol out of these arrays seems easy enough but lots of rapid cognitive and motor operations have to be done — user holds communicative intent in WM; tunes out irrelevant sensory information; scans grid visually, locates target, maps physical movement; touches screen. If small delays creep into this process over and over message generation slows down. In noisy social or educational settings micro-delays can shut someone out completely from conversations. Research has shown that changing how the display itself is laid out (spatial arrangement, grouping symbols by color, crowding control) directly affects how quickly a user can find and select a target (e.g., Jones et al., 2014). For autistic participants specifically changing how a simulated display was laid out resulted in measurable differences in eye-gaze fixation as well as overall touch latency. More recent work shows visual fixation speed correlates with physical touch latency too so selection speed is concrete outcome measure. But real world communication doesn't happen in a vacuum. An adolescent using an AAC device has classmates talking, TV playing in the background or people walking across their field of view — noise and visual chaos everywhere. General cognitive literature says distractors slow response times down and multisensory processing profiles in autism are often very different from unimodal ones but sensory-processing findings

haven't been applied to structured AAC testing yet. Studies almost always change the interface itself so if we keep the AAC grid exactly the same how does noise and visual chaos from the room affect speed when an autistic teenager selects a known symbol? This study isolates that variable. Everything else is held constant (grid layout, symbols, item positions, tablet hardware, viewing distance, touch mechanics). Then we can experiment on impact of external sensory competition alone. We will compare three conditions: low distraction baseline; auditory competition setup; combined audiovisual competition setup. This design cleanly separates environmental sensory load from interface engineering and reveals everyday communication barrier that AAC research has largely ignored.

Review of Existing Literature

A huge body of AAC literature shows visual perceptual decisions determine how fast a user scans a screen. Light et al. showed many visual perceptual decisions were made during scanning. If we use this technique in AAC we can isolate what actually matters. A lot of AAC studies change the interface itself (grid layout, symbols, item positions, tablet hardware, viewing distance, touch mechanics) so if we keep the AAC grid exactly the same how does noise and visual chaos [12] showed that speed of scanning depends on whether a user sees an image of the symbol before it appears on screen or not. Many studies show visual decision making Visual percepts are important for scanning a screen quickly [1]. Multisensory processing profile in autism also diverges ... reviewed evidence spatial layouts and symbol clustering speed search and touch times on grids while poorly planned visual details distract unnecessarily (Wilkinson & Madel tested these dynamics directly: autistic individuals and Down syndrome individuals

fixated targets faster when symbols with matching internal colors were clustered together and moved mouse quicker when they were too (this kind of clustering also seemed to shield users from irrelevant visual distractors) just as earlier work showed structured perceptual layouts sped up visual search tasks for meaningful symbols). Taken together these studies show selection latency is very sensitive to cognitive demands within boundaries of screen. More recently people have connected those early visual processes directly to physical execution: Wilkinson et al found initial visual fixation pathways predicted final motor-selection times in autistic individuals very closely. This is what our current proposal is all about: it shows raw selection latency is valid measure of efficiency (not special eye tracking infrastructure). Another line of work is how autistic people process crowded sensory environments. Ostrolenk et al. found reduced multisensory facilitation typical in adolescents and adults with autism on reaction-time tests of basic tones and flashes. Developmental work shows multisensory interactions are there in autistic teenagers but processing patterns vary wildly across task and individual participant. Developmental work at larger scale is Downing et al., who show modality of the distractor changes reaction time (dual audiovisual distractors impose heavy cognitive load during active target discrimination). Gap in literature is where these two fields meet: AAC researchers study interface design; sensory psychologists track reaction times on abstract laboratory tasks. We bridge that divide by testing performance on a functional realistic AAC display (interface left intact) only manipulating the external environment. This is a direct unconfounded test of whether everyday auditory and visual distractions slow down access to symbols for autistic adolescents.

Research Questions & Hypotheses

The central question of this study is: Does external sensory competition (auditory and audiovisual distractors relative to quiet baseline) speed up or slow down symbol-selection latency on a fixed unchanging AAC grid display for autistic adolescent users? The hypothesis is that mean symbol-selection speeds will differ significantly between the three sensory environments. That is, how much do they shift (not just assuming multi-modal distraction will impair performance uniformly for every single participant). To get a better picture we will also track 3 secondary questions: Do competing environmental inputs affect final symbol-selection accuracy? Are these environmental effects stable across cohort (or do they show distinct patterns of vulnerability)? Does performance change across trials in a condition (i.e. fast adaptation, cumulative fatigue or practice effects).

Methodology

Experimental Design and Participant Selection: We use repeated measures within participants designs so that everyone does all three sensory testing blocks. Because autistic people are so heterogeneous this is a very strong control mechanism: each teenager is their own baseline (no confounds from baseline motor speed, language comprehension, visual acuity or prior AAC experience). We plan to recruit a pilot sample (~24 autistic adolescents [12–18 years old] through clinics near us, specialized schools in the community, advocacy groups and family networks). This is exploratory pilot study so we will report effect size estimates and 95% CI for all analyses since future confirmatory studies will need to be powered correctly. People must meet inclusion criteria below: Clinical or educational diagnosis of autism documented in records. Experience with direct-touch grid based AAC (functional ability to do a brief practice trial); normal or

corrected vision and hearing so person can engage fully with task. Adolescents who cannot reliably use direct touch or whose motor and sensory profiles don't match up with a standard screen-touch task will be supported with standard accommodations outside the study but not enrolled in this experiment. Selection latency is our main metric so we need some baseline of physical access before valid data can be collected.

Apparatus and invariant AAC interface

Hardware artifacts that could bias results: All testing sessions will have identical tablet models, screen orientations, brightness levels, stand angles, viewing distances and touch input thresholds so as not to confound the results. The experimental AAC layout will consist of a static set of highly imageable familiar symbols arranged in a grid (note: these symbols are not moving around; they are completely anchored in exactly the same positions throughout the entire project — no resizing, changing color schemes, animating or swapping labels between blocks). By keeping the interface invariant any changes observed are due only to room environment. Before timed trials each participant goes through familiarization phase to make sure they recognize the symbols (to minimize learning curves while collecting active data). Every participant goes through three matched testing blocks (that are common in everyday life) (Condition A: Low-distraction baseline - Participant interacts with invariant grid inside quiet neutral testing room. Task-irrelevant media, background noise and visual clutter are kept to an absolute minimum); Condition B (Auditory Competition): The task is interrupted by loud sounds at random times during the experiment ("click"). Condition C (Audiovisual Competition): Same interface but now the participant has auditory and visual environmental distractors at once (visual

competition) but uses exactly the same background audio track from Condition B playing with an unrelated video clip (this video will be shown on a separate standard screen in the periphery of the field of view [e.g., cluttered living room or active classroom]). For strict experimental control, all external factors (decibel levels, screen position, etc.) will be standardized and logged; stimuli will be age-appropriate, non-startling, and pre-tested during pilot phase for tolerability.

Procedural flow and trial mechanics

Specific trial mechanics are based on automated nonverbal pacing: at the start of each trial, the target cue appears on screen to brief the participant before the response grid appears (no further auditory prompts during timed selection window); after short delay invariant grid appears and the participant touches symbol. Presentation software records selection latency (ms from moment of grid onset to screen touch registered automatically). Accuracy is also recorded then as a binary correct/incorrect outcome (not cleaning data after seeing final results). Trials where accidental touches or technical glitches occurred will be flagged and excluded explicitly (predefined data rules). Blocks have the same number of trials and targets; targets are completely randomized within blocks to avoid anticipation, but order of presentation of conditions themselves (not a target) is counterbalanced across participants in a Latin-square schedule so no practice or carryover effects could occur. Breaks between blocks reduce fatigue. The whole session runs before a short practice run, which does not enter final analysis. If a participant cries or leaves at any point in the session, it ends immediately

Statistical framework and analytical approach

The main focus of our analysis will be on symbol selection latency for successful trials: before running inferential tests, we check skewness and extreme outliers of latencies according to our pre-specified data rules. Main statistical analysis will be done with linear mixed-effects model sensory condition as fixed effect individual as random intercept Trial numbers and condition order will be planned covariates in the model Raw latency data if non-normal log transform or transition into GLM framework (appropriate generalized linear mixed-model) Planned contrasts will be run to systematically compare: Auditory competition with quiet baseline Audiovisual competition with quiet baseline Audiovisual competition directly against auditory competition All reported \[p\]-values will be accompanied by effect sizes and 95% CI (if available) so magnitude can be seen clearly Accuracy of subject performance will be assessed separately with a mixed effects logistic regression model. Participant-level data plots will be produced alongside aggregate scores so clinically meaningful heterogeneity isn't masked behind broad group means

Rigor Experimental Controls Feasibility

The strength of this design is that the AAC interface remains unchanged. Hardware models, grid layouts, symbol coordinates, target frequencies, physical access methods, and spatial viewing geometries are all the same for all three test conditions. A scripted uniform protocol standardizes investigator instructions, while software timestamps are used to eliminate human stopwatch error completely (random session logs and screen recordings are checked against automated data files to ensure timing accuracy and condition coding). It's very practical because tablets from commercial vendors are used (external speakers), standard presentation software; plus it can be

done without disrupting or changing a participant's highly personalized everyday communication vocabulary.

Ethical Protection of the Participant

Because this study works with minors who may have complex communication needs, we have very strict ethical protections in place: Institutional review board approval before any interaction with participants; parents or guardians give written consent (AAC support where possible); adolescents assent via accessible plain-language documentation (AAC support if needed); participation is entirely voluntary (withdraw anytime without penalty or loss of outside services). Volumes of sensory elements and visual elements are kept to safe everyday levels and stopped immediately if someone is distressed or wants to stop. Structurally, the protocol doesn't frame autistic sensory differences as deficits - it wants to understand how specific environmental designs can be optimized so communication access is reliable.

The Expected Results Scope and Possible Structural Limitations

This study wants to know if external sensory contexts influence selection latency when the digital interface is perfectly stable. If we get a significant condition effect, then AAC efficiency is context-dependent (so traditional assessments done only in pristine quiet rooms may not measure what an individual actually does in real life), or a null finding would say that a well-learned invariant grid layout has strong cognitive stability resistant to a very large amount of external environmental chaos. Either way, we have empirical evidence for classroom design, clinical

evaluation protocols, and adaptive interface systems. Of course there are limitations too. This is exploratory pilot work with a small sample size, so can't be representative of autistic AAC users; lab distractors are artificial; the structured target-matching task cannot represent the organic cognitive load of spontaneous self-generated conversation, etc. So this project doesn't make sweeping generalizations about the whole autism population or assistive technology contexts. It sets the stage for future research (which might eventually move this invariant paradigm into naturalistic dialogues, personalized sensory profiles, eye-tracking analytics, physiological monitoring, or context-aware adaptive AAC software).

Creativity/Intellectual Contribution

The conceptual novelty of the project is that it flips a traditional research question on its head: instead of asking what AAC interface layout works best, we ask how stable is a user's access to the same layout when the surrounding sensory environment varies? Adolescents with high-tech AAC communicate in loud, visually busy, and socially noisy environments. But environmental sensory competition is rarely separated as an independent experimental variable while the interface is held perfectly constant. This study provides a single tightly controlled, very reproducible framework that is cost-effective. Its innovation is not that we made the assistive device more complex but that we experimentally separated device architecture from external environmental load. Ultimately, these results will give clear design principles for future software engineers but also concrete evidence to clinicians, families, and educators as well, striving to ensure reliable communication access across real-world settings for these individuals.

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